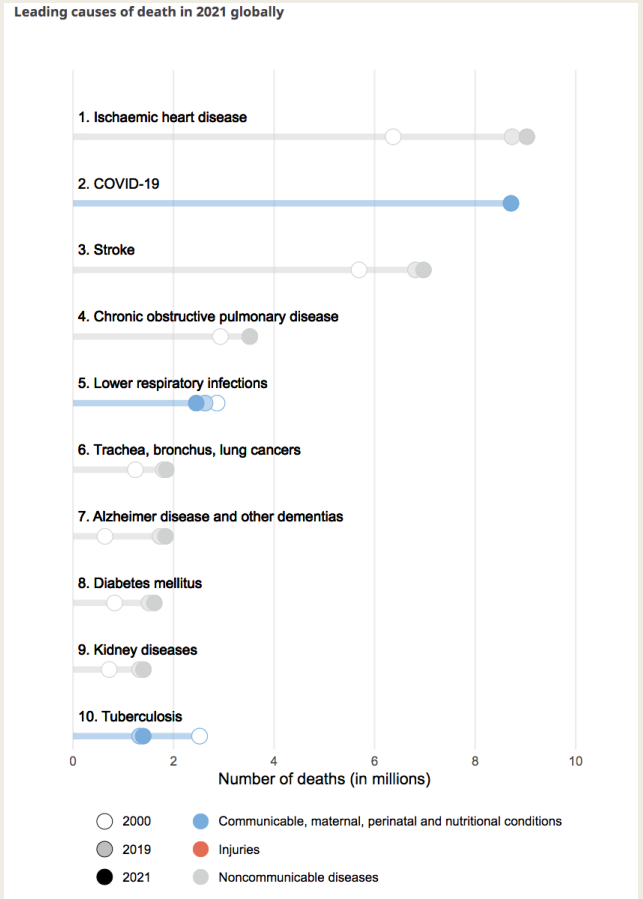
EXERCISE PRINCIPLES: PHYSICAL ACTIVITY AND HEALTH PROMOTION

Bryan Tin
Specialist Respiratory Physiotherapist
Royal Brompton Hospital – Guy's and St. Thomas' NHS Foundation Trust

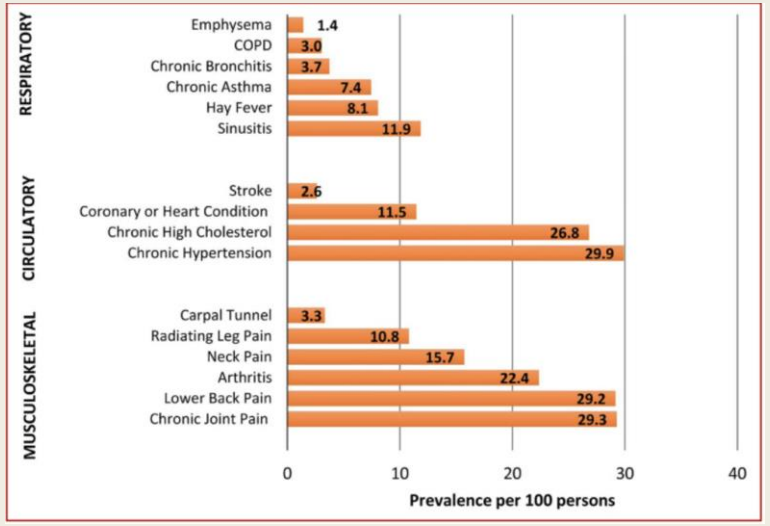
Learning Objectives

- Understanding the current state of physical activity and sedentary behaviour
- Understanding the benefits of physical activity
- Brief view of the current physical activity guidelines along with its evidence
- Understanding tools for physical activity monitoring
- Understanding the barriers to physical activities

What is killing us?







"In view of the prevalence, global reach and health effect of physical inactivity, the issue should be appropriately described as Pandemic, with far-reaching health, economic, environmental and social consequences" – The LANCET Physical Activity Series 2012

Where are we at currently?

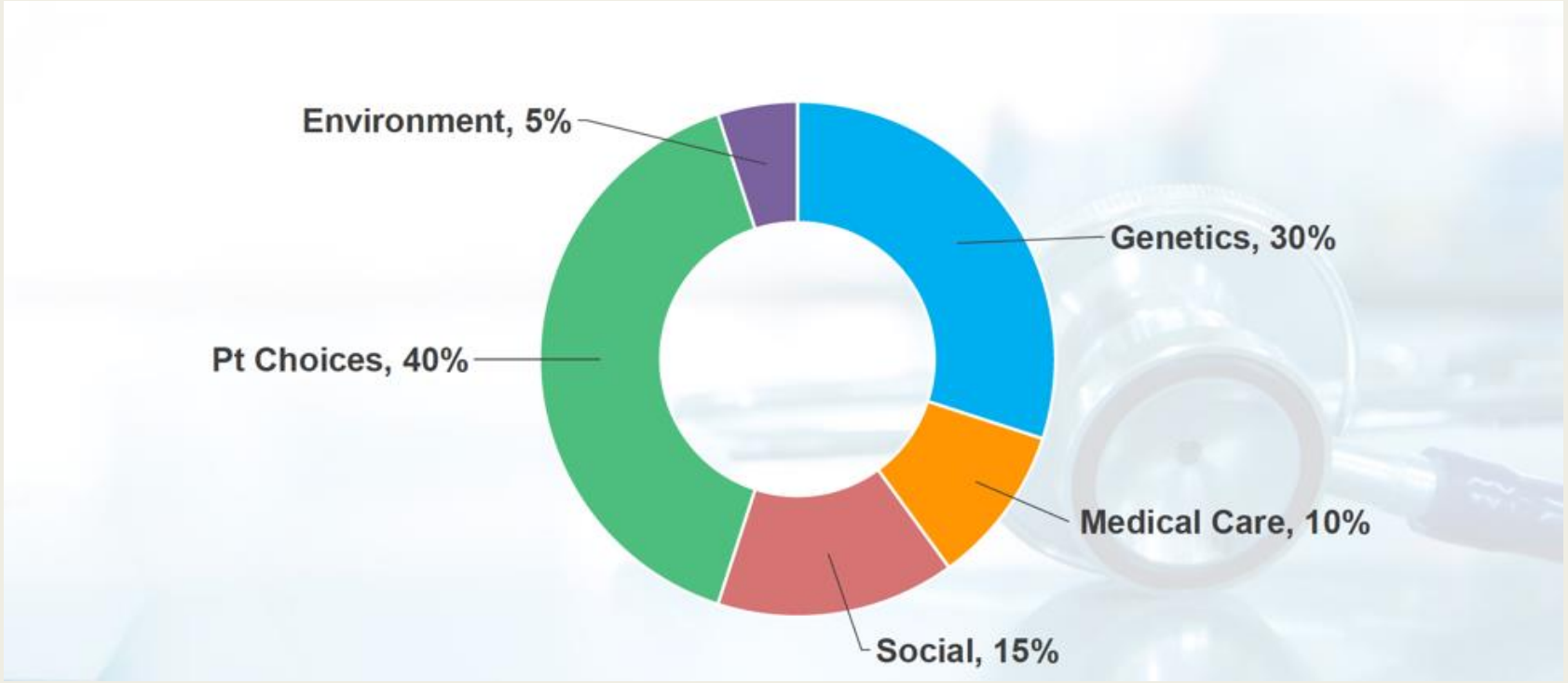
- **Global pandemic** of physical inactivity
  - *Over 5 million deaths per year*
  - *Over 67.5 billion of economic burden per year (53.8 billion in 2013)*
- More than 1.4 billion adults are not achieving recommended physical activity level
- Medical advancement largely exposed risks of physical inactivity
- Significant shift in activity patterns post COVID-19
  - *Work*
  - *Daily activities e.g. shopping*

If we dive deeper into the finances…

- Health care related cost is 53.8 billion in 2013
- Disease distribution
  - *CHD – 5 billion*
  - *Stroke – 6 billion*
  - *T2DM – 37.6 billion*
  - *Breast cancer – 2.7 billion*
  - *Colon cancer 2.5 billion*
- By region
  - *Europe - 11.7 billion (0.55%)*
  - *Africa – 0.6 billion (0.33%)*
  - *Latin America and Caribbean – 3.2 billion (0.53%)*
  - *North America – 25.7 billion (0.84%)*
  - *South East Asia – 0.9 billion (0.22%)*
  - *Western Pacific region – 9.2 billion (0.54%)*
- Physical inactivity related deaths cost 13.7 billion in productivity loss

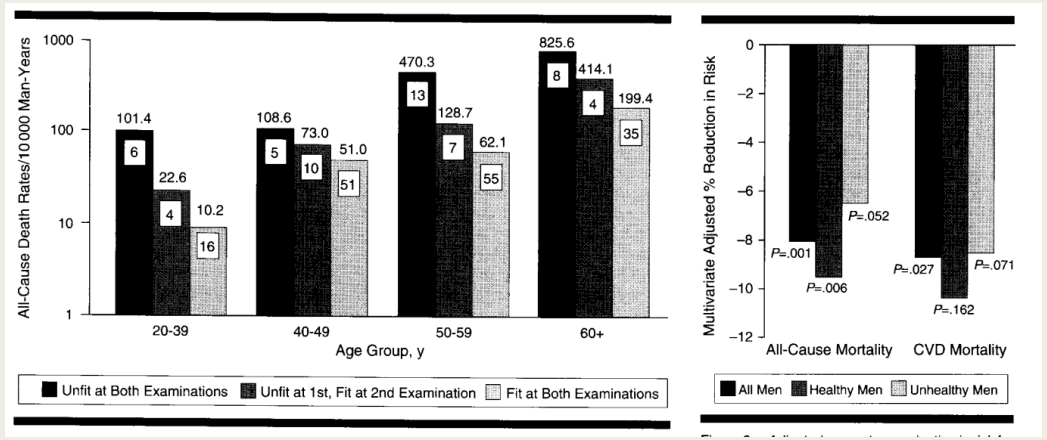
THIS IS MORE THAN 10 YEARS AGO!!!

The power of health behaviours



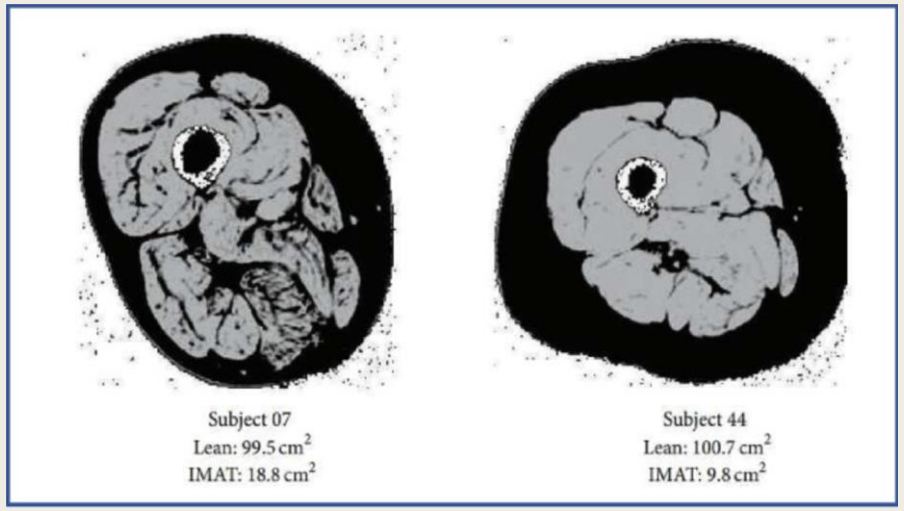
Benefits of physical activity

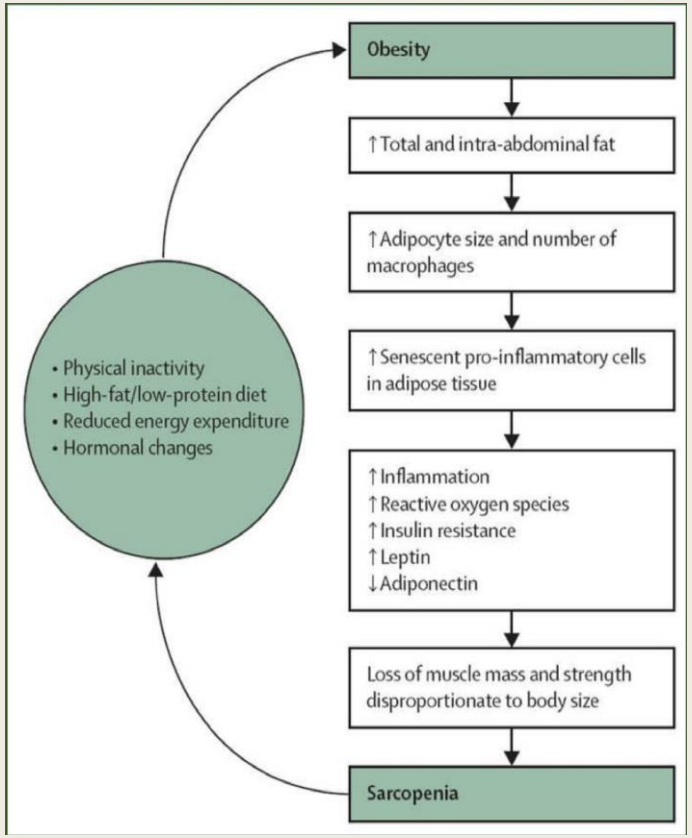
- Prevention and treatment of non-communicable disease
- Reduce stress and anxiety
- Mitigate risks and prevent obesity
- Improve bone health and prevent fractures
- Prevent and treat heart disease
- Lower risk of breast and colon cancer
- Reduce risk of dementia



Skeletal health

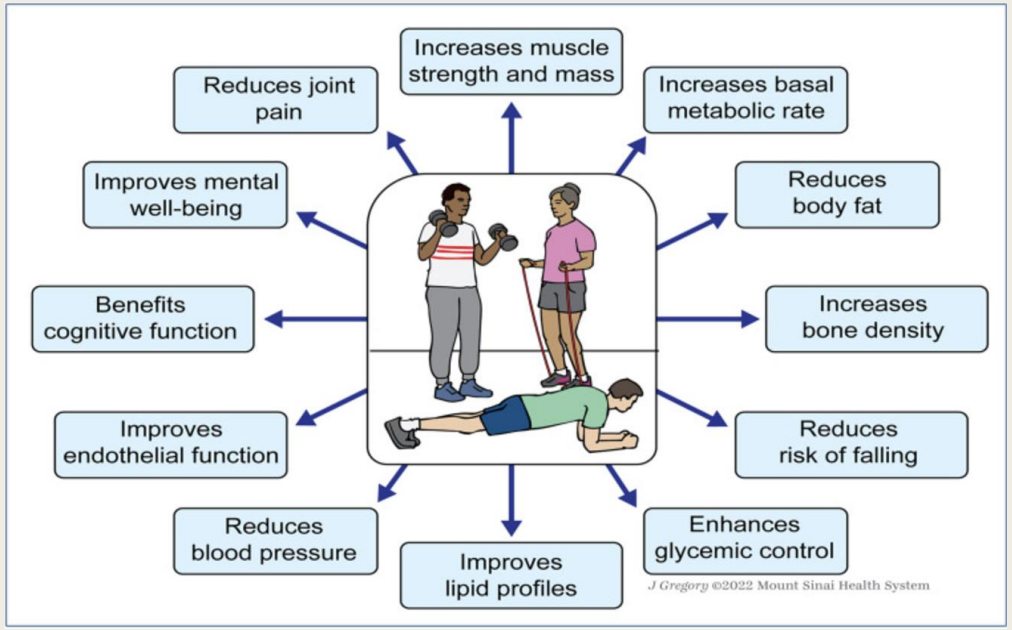
- Common physical inactivity-related skeletal outcomes include:
  - *Sarcopenia*
  - *Loss of muscle strength and power*
  - *Decrease in muscle flexibility*
  - *Worsening balance*
  - *Decline in basal metabolic rate*
  - *Increase in fatt mass*
  - *Decrease in bone mineral density*
- Muscle mass decreases by 3-8% per decade after age 30 with a greater decline after age 60
  - *In the absence of resistance training*
- Accelerated the process of fat infiltration into skeletal muscles
- Physical inactivity is found to be associated with narrower lumbar intervertebral discs

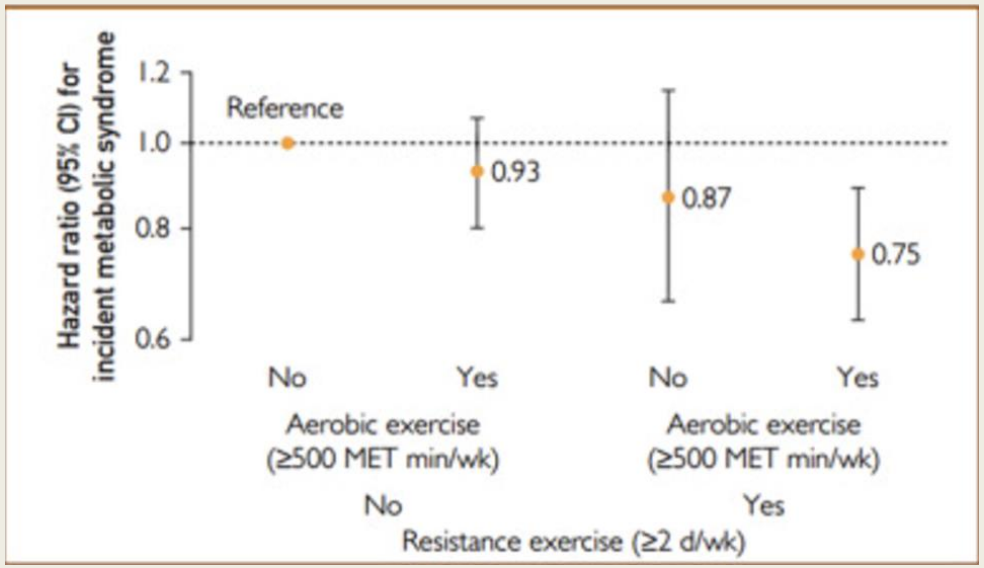




Skeletal health

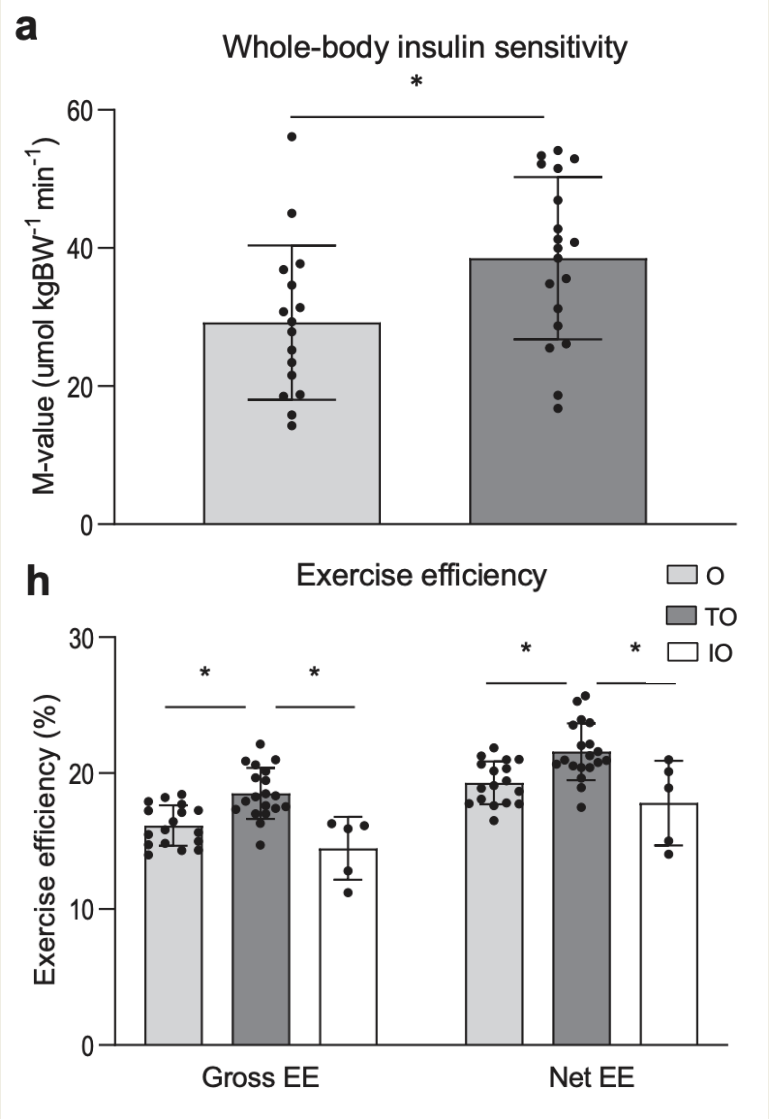
- Physical activity and exercise helps to
  - *Muscle fibre hypertrophy*
  - *Improve neuromuscular control*
  - *Strength gaining*
  - *Conversion to endurance type fibres*
  - *Improve stability provided to joints*
  - *Other metabolic benefits*





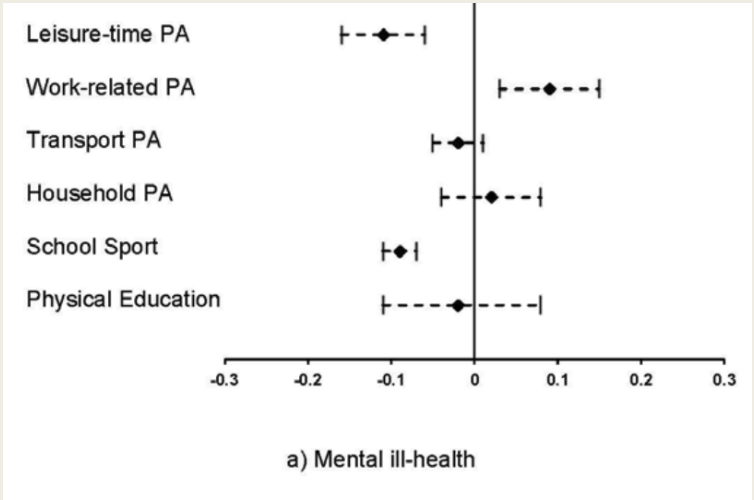
Benefits in skeletal health

- Observational studies suggested 11-40% decline in fracture risks
- 15% difference in energy cost while comparing physically-active older adults to the sedentary counterparts
- Improvement in recovery following gait perturbations with physical activities
- Evidence suggested high level of physical activity with low sedentary behaviour significantly associated with stronger muscle strength and power
  - *Specifically with the lower body and hand grip*
    - Highlighted the need of these being included in health screen outcome measures?
    - Lower body strength and power significant associates with ageing
  - *But ultimately all levels of physical activities are positively associated with outcome*
  - *Also showed greater ability in living independently*
- Exercise training (avg. >12500 steps per day) induces mitochondria content and mitigate age-related mitochondrial function decline



Benefits in mental health and cognition

- Physical activity has been well recognised as the prevention and management of mental illness, including
  - *Mental disorder such as depression and anxiety*
  - *Promotion of mental health such as wellbeing*
  - *Cognitive functioning*
- Leisure time physical activity has shown to have a stronger relationship with mental health
  - *Domains of physical activity has various outcomes on mental health*
- Strong association between minimum 4 years of recreational/competitive sport and better mental health score
- Evidence showed trend of reduced neuropsychiatric symptoms in dementia patients



Benefits in mental health and cognition

- Key mechanisms
  - *Enhance self-efficacy*
  - *Expose individual to challenges*
  - *Offering opportunities to develop confidence and sense of mastery*
  - *Increased social interactions*
  - *Diversion of attention from stressful life events*
  - *Higher likelihood to be motivated*
  - *Creates satisfaction of psychological needs*
  - *Improvement in sleep pattern*

Benefits in mental health and cognition

- Physiological changes
  - *Release of endorphins which eases pain and produce the feeling of euphoria*
  - *Release of dopamine, noradrenaline and serotonin, creating a "feeling good" sensation*
  - *Experiment showed decreased fMRI activation of right anterior prefrontal cortex in children, mirror a more mature adult*
  - *Increase neuroplastic capacity*
  - *Growth of new neuromuscular pathway*
  - *Potentiation of neurotransmitter systems*
  - *Enhanced neurotrophic growth factors to facilitate neurogenesis and angiogenesis*
  - *Enhanced connectivity within functional neuroanatomical circuitary*

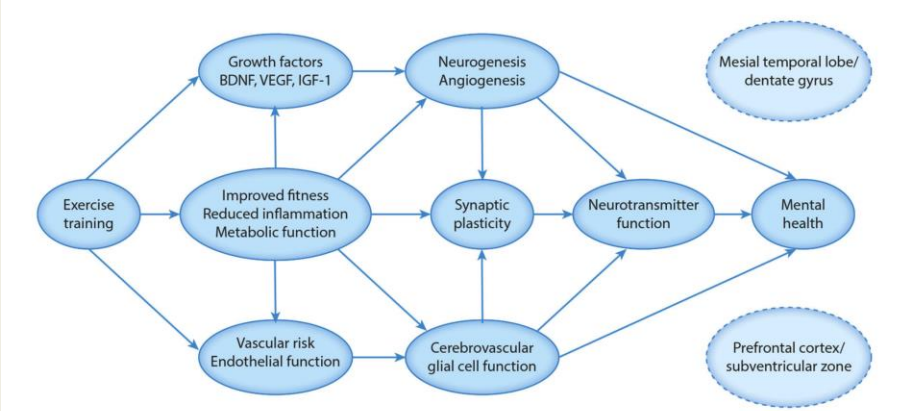
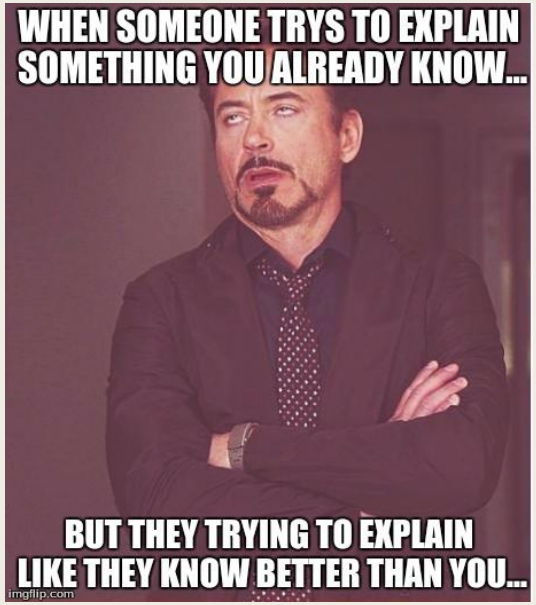
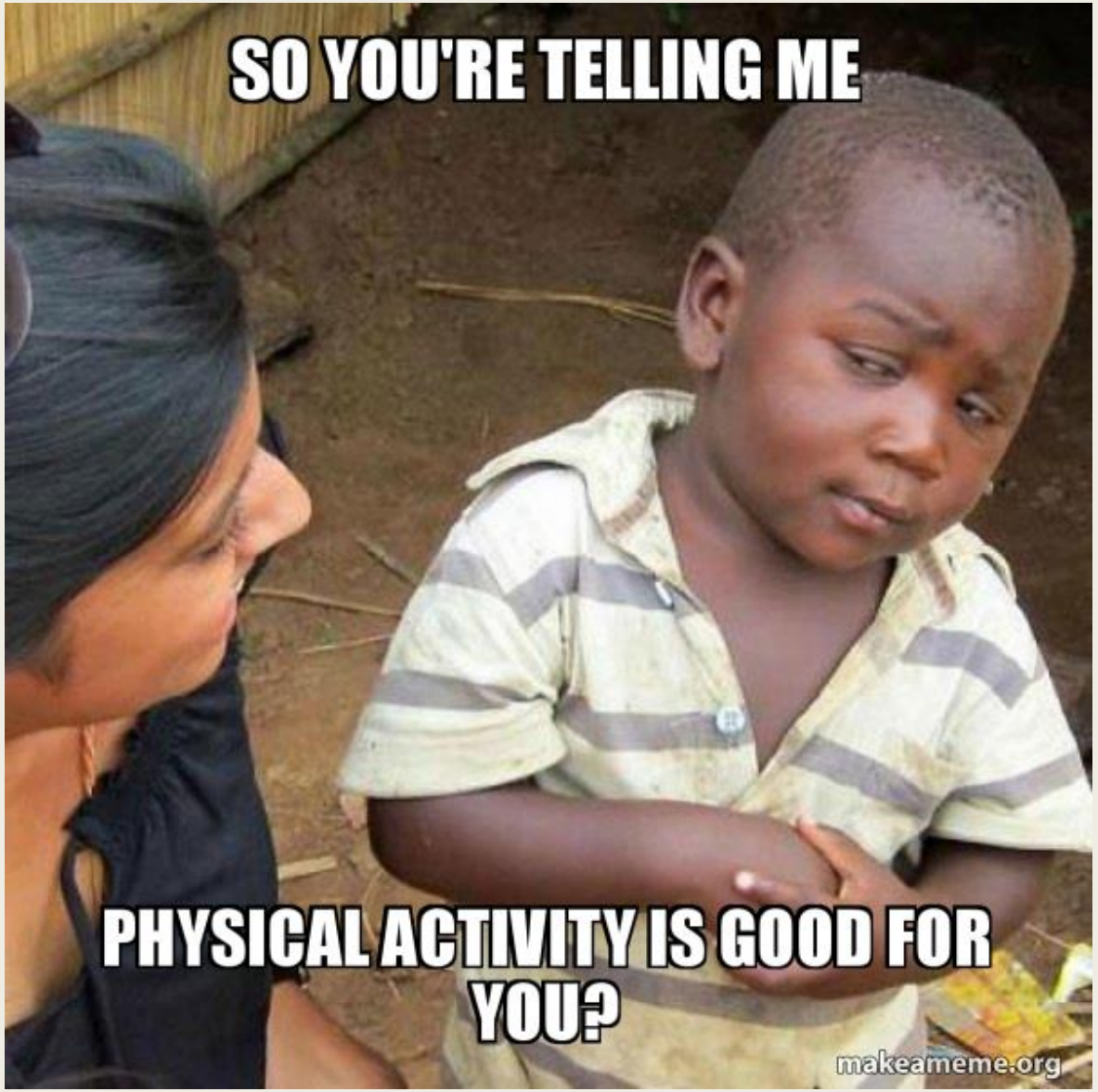


Figure 2.
Conceptual model of neurobiological mechanisms by which exercise training improves mental health outcomes. Abbreviations: BDNF, brain-derived neurotrophic factor; IGF-1, insulin-like growth factor 1; VEGF, vascular endothelial growth factor.

Benefits in cardiorespiratory health

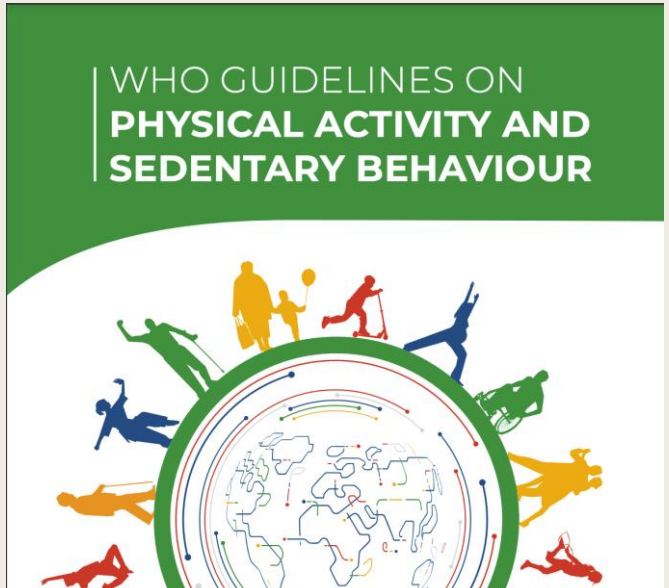
YOU SHOULD ALREADY KNOW!




SO YOU'RE TELLING ME
PHYSICAL ACTIVITY IS GOOD FOR YOU?
makeameme.org

Current guidelines

- WHO updated guidelines on physical activity and sedentary behaviour in 2020
  - *Targeting all children, adults, older adults and pregnant individuals*
  - *Included recommendations to limit sedentary behaviours*
  - *Approach of little is better than nothing*



Current guidelines

- What's new:
  - *Removal of previous stipulation that physical activities should be accumulated in at least 10 minutes bouts*
  - *Now specifies a target range for accumulated duration, in comparison to previous minimal threshold*
  - *Multicomponent activities emphasising on functional balance and strength training now applies to all older adults, instead of just those with poor mobility*
  - *Included sedentary limitation*

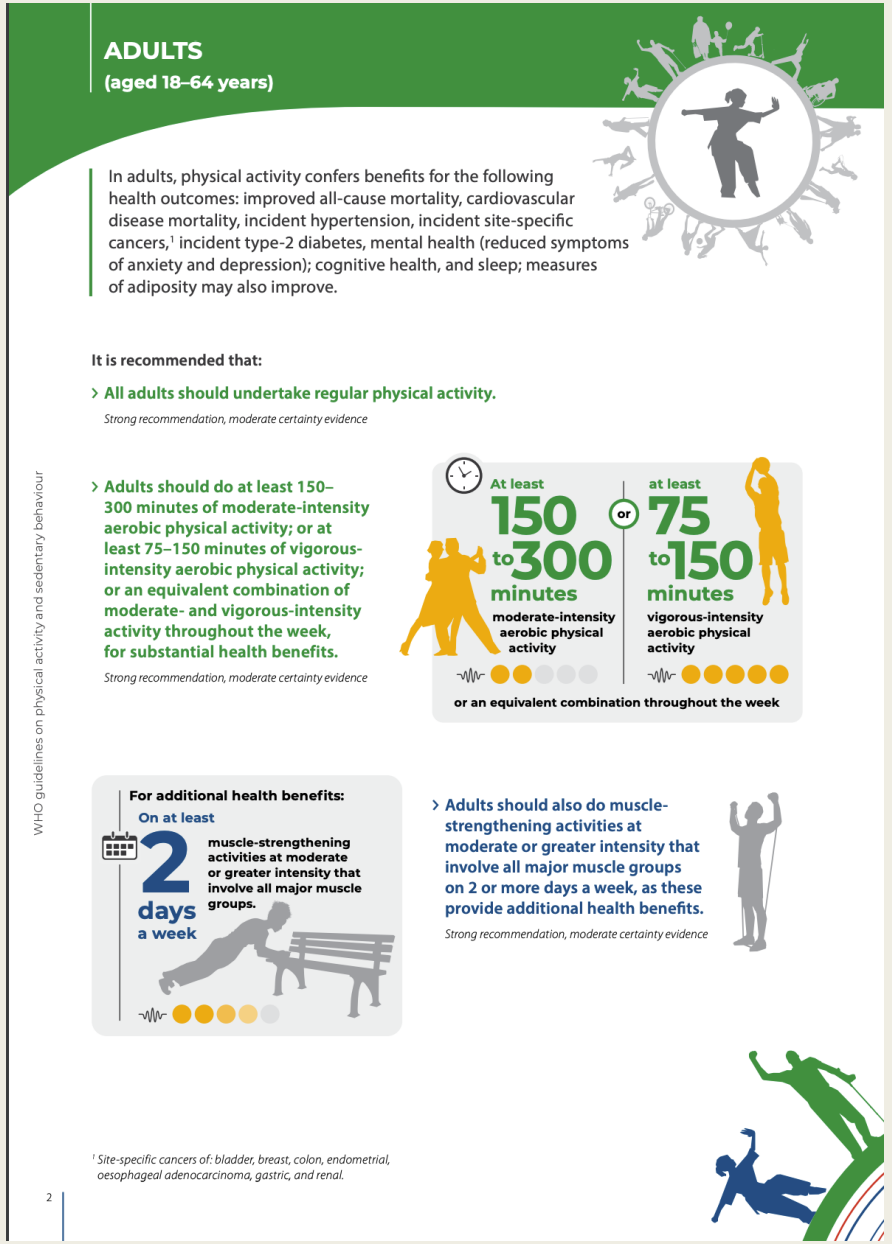
Current guidelines

- Age 5 – 17
- In terms of physical activity, it is recommended that:
  - *At least an average of 60min/day of moderate-to-vigorous intensity, mostly aerobic, physical activity, across the week;*
  - *Vigorous-intensity aerobic activities, as well as those that strengthen muscle and bone should be incorporated at least 3 days a week.*
- In terms of sedentary behaviour, It is recommended that:
  - *Limit the amount of time spent being sedentary, particularly the amount of recreational screen time.*



Current guidelines

- Age 18 -64
- In terms of physical activity, It is recommended that:
  - *All adults should undertake regular physical activity;*
  - *Adults should do at least 150–300 min of moderate-intensity aerobic physical activity, or at least 75–150min of vigorous-intensity aerobic physical activity, or an equivalent combination of moderate-intensity and vigorous-intensity activity throughout the week for substantial health benefits;*
  - *Adults should also do muscle-strengthening activities at moderate or greater intensity that involve all major muscle groups on 2 or more days a week, as these provide additional health benefits.*
  - *Adults may increase moderate-intensity aerobic physical activity to >300min, or do >150min of vigorous-intensity aerobic physical activity, or an equivalent combination of moderate-intensity and vigorous-intensity activity throughout the week for additional health benefits*



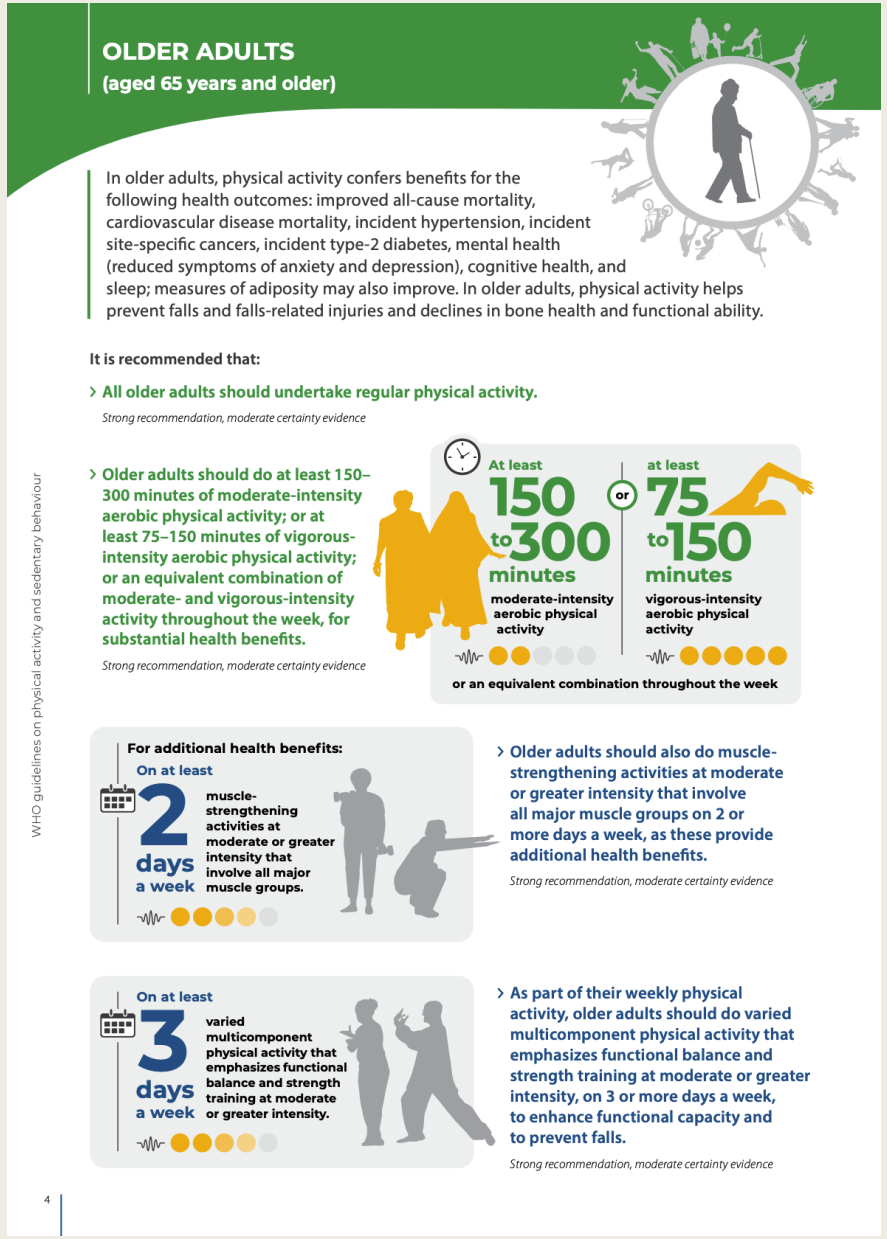
Current guidelines

- Age 18– 64
- In terms of sedentary behaviours, It is recommended that:
  - *Adults should limit the amount of time spent being sedentary.*
  - *Replacing sedentary time with physical activity of any intensity (including light intensity) provides health benefits;*
  - *To help reduce the detrimental effects of high levels of sedentary behaviour on health, adults should aim to do more than the recommended levels of moderate-to vigorous physical activity.*



Current guidelines

- Older adults (65 or above)
- In terms of physical activity, It is recommended that:
  - *As for adults*
  - *As part of their weekly physical activity, older adults should do varied multicomponent physical activity that emphasises functional balance and strength training at moderate or greater intensity on 3 or more days a week, to enhance functional capacity and to prevent falls*
- In terms of sedentary behaviours, It is recommended that:
  - *As for adults*



Current guidelines

- Pregnant and postpartum women
- In terms of physical activity, It is recommended that all pregnant and postpartum women without contraindication should:
  - *Undertake regular physical activity throughout pregnancy and postpartum;*
  - *Do at least 150min of moderate-intensity aerobic physical activity throughout the week for substantial health benefits;*
  - *Incorporate a variety of aerobic and muscle-strengthening activities. Adding gentle stretching may also be beneficial.*
  - *Women who, before pregnancy, habitually engaged in vigorous-intensity aerobic activity or who were physically active can continue these activities during pregnancy and the postpartum period.*
- In terms of sedentary behaviours,
  - *Pregnant and postpartum women should limit the amount of time spent being sedentary. Replacing sedentary time with physical activity of any intensity (including light intensity) provides health benefits.*



Current guidelines



ADULTS AND OLDER ADULTS WITH CHRONIC CONDITIONS (aged 18 years and older)

Physical activity can confer health benefits for adults and older adults living with the following chronic conditions: **for cancer survivors** – physical activity improves all-cause mortality, cancer-specific mortality, and risk of cancer recurrence or second primary cancer; **for people living with hypertension** – physical activity improves cardiovascular disease mortality, disease progression, physical function, health-related quality of life; **for people living with type-2 diabetes** – physical activity reduces rates of mortality from cardiovascular disease and indicators disease progression; and **for people living with HIV** – physical activity can improve physical fitness and mental health (reduced symptoms of anxiety and depression), and does not adversely affect disease progression (CD4 count and viral load) or body composition.

It is recommended that:

› **All adults and older adults with the above chronic conditions should undertake regular physical activity.** *Strong recommendation, moderate certainty evidence*

At least **150 to 300 minutes** moderate-intensity aerobic physical activity

or

at least **75 to 150 minutes** vigorous-intensity aerobic physical activity

or an equivalent combination throughout the week

› **Adults and older adults with these chronic conditions should do at least 150–300 minutes of moderate-intensity aerobic physical activity; or at least 75–150 minutes of vigorous-intensity aerobic physical activity; or an equivalent combination of moderate- and vigorous-intensity activity throughout the week for substantial health benefits.**

Strong recommendation, moderate certainty evidence

› **Adults and older adults with these chronic conditions should also do muscle-strengthening activities at moderate or greater intensity that involve all major muscle groups on 2 or more days a week, as these provide additional benefits.**

Strong recommendation, moderate certainty evidence

For additional health benefits:

On at least **2 days a week** muscle-strengthening activities at moderate or greater intensity that involve all major muscle groups.

On at least **3 days a week** varied multicomponent physical activity that emphasizes functional balance and strength training at moderate or greater intensity.

› **As part of their weekly physical activity, older adults with these chronic conditions should do varied multicomponent physical activity that emphasizes functional balance and strength training at moderate or greater intensity on 3 or more days a week, to enhance functional capacity and prevent falls.**

Strong recommendation, moderate certainty evidence

WHO guidelines on physical activity and sedentary behaviour

8



ADULTS AND OLDER ADULTS WITH CHRONIC CONDITIONS (aged 18 years and older)

For additional health benefits:

more than **300 minutes** moderate-intensity aerobic physical activity

or

more than **150 minutes** vigorous-intensity aerobic physical activity

or an equivalent combination throughout the week

› **When not contraindicated, adults and older adults with these chronic conditions may increase moderate-intensity aerobic physical activity to more than 300 minutes; or do more than 150 minutes of vigorous-intensity aerobic physical activity; or an equivalent combination of moderate- and vigorous-intensity activity throughout the week for additional health benefits.**

Conditional recommendation, moderate certainty evidence

GOOD PRACTICE STATEMENTS

- When not able to meet the above recommendations, adults with these chronic conditions should aim to engage in physical activity according to their abilities.
- Adults with these chronic conditions should start by doing small amounts of physical activity and gradually increase the frequency, intensity and duration over time.
- Adults with these chronic conditions may wish to consult with a physical activity specialist or health-care professional for advice on the types and amounts of activity appropriate for their individual needs, abilities, functional limitations/complications, medications, and overall treatment plan.
- Pre-exercise medical clearance is generally unnecessary for individuals without contraindications prior to beginning light- or moderate-intensity physical activity not exceeding the demands of brisk walking or everyday living.

In adults, including cancer survivors and people living with hypertension, type-2 diabetes and HIV, higher amounts of sedentary behaviour are associated with the following poor health outcomes: all-cause mortality, cardiovascular disease mortality and cancer mortality, and incidence of cardiovascular disease, cancer and incidence of type-2 diabetes.

For cancer survivors, and adults living with hypertension, type-2 diabetes and HIV, it is recommended that:

› **Adults and older adults with chronic conditions should limit the amount of time spent being sedentary. Replacing sedentary time with physical activity of any intensity (including light intensity) provides health benefits.**

Strong recommendation, low certainty evidence

› **To help reduce the detrimental effects of high levels of sedentary behaviour on health, adults and older adults with chronic conditions should aim to do more than the recommended levels of moderate- to vigorous-intensity physical activity.**

Strong recommendation, low certainty evidence

LIMIT the amount of time spent being sedentary

REPLACE with more physical activity of any intensity (including light intensity).

Executive summary

9

Current guidelines

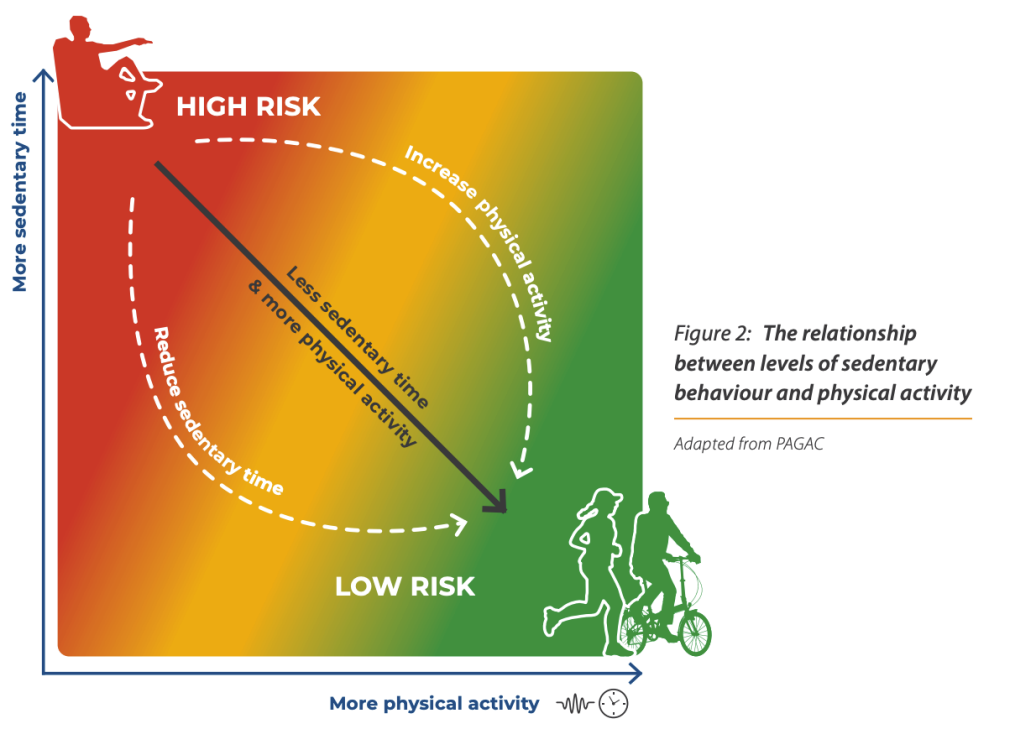


Figure 2: ***The relationship between levels of sedentary behaviour and physical activity***

Adapted from PAGAC

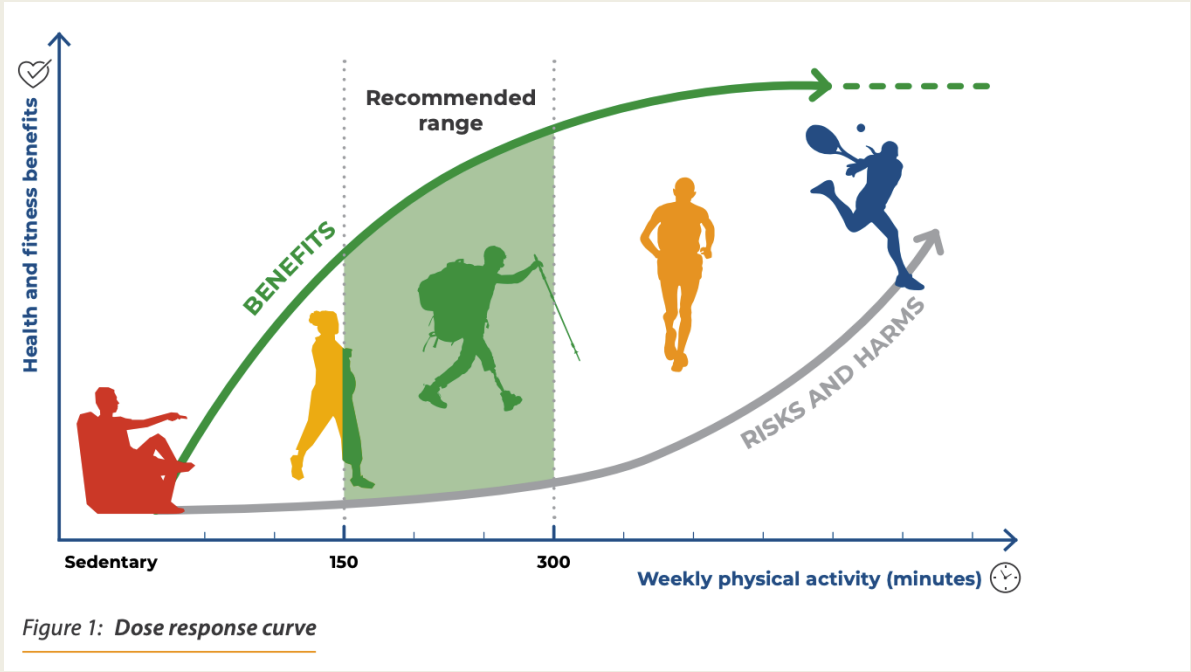
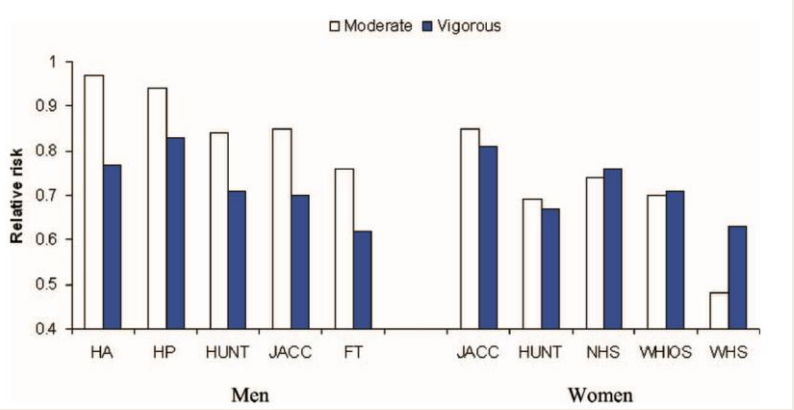


Figure 1: ***Dose response curve***

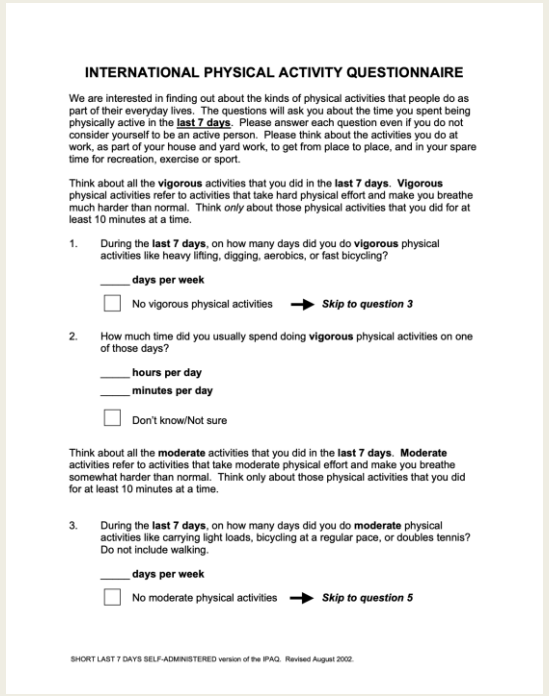
Evidence of current guidelines

- Physical activity in any bouts of length is associated with improved health outcomes, including all-cause mortality
- 150 min of moderate-intensity exercise per week may improve physical and mental quality of life, and cognitive function in those aged 45–50 years and older
- Certain ranges of physical activity accumulated duration captures the maximal risk reduction of health outcomes, with conclusions that going beyond this range does not further reduce the risk of major outcomes
- Cross-sectional studies and exercise interventions suggest that these doses of activity are associated with favourable changes in blood pressure lipid and lipoprotein profiles, markers of inflammation, insulin sensitivity and other risk factors for chronic diseases
- Recommended doses of activity are associated with 20–30% reductions in the risks of cardiovascular disease, type 2 diabetes, depression, cognitive impairment in older adults, post-menopausal breast cancer, and all-cause. Higher doses of activity may be necessary to reduce the risks of other common cancers, such as colon cancer and, possibly, high-grade or advanced prostate cancer



Outcome measures

- Self-reported questionnaires
- Logs and diaries
  - *Accuracies of recall*
- Pedometers
- Accelerometers
- Heart rate monitor



INTERNATIONAL PHYSICAL ACTIVITY QUESTIONNAIRE

We are interested in finding out about the kinds of physical activities that people do as part of their everyday lives. The questions will ask you about the time you spent being physically active in the **last 7 days**. Please answer each question even if you do not consider yourself to be an active person. Please think about the activities you do at work, as part of your house and yard work, to get from place to place, and in your spare time for recreation, exercise or sport.

Think about all the **vigorous** activities that you did in the **last 7 days**. **Vigorous** physical activities refer to activities that take hard physical effort and make you breathe much harder than normal. Think *only* about those physical activities that you did for at least 10 minutes at a time.

1. During the **last 7 days**, on how many days did you do **vigorous** physical activities like heavy lifting, digging, aerobics, or fast bicycling?

   ____ **days per week**

   ☐ No vigorous physical activities → ***Skip to question 3***

2. How much time did you usually spend doing **vigorous** physical activities on one of those days?

   ____ **hours per day**

   ____ **minutes per day**

   ☐ Don't know/Not sure

Think about all the **moderate** activities that you did in the **last 7 days**. **Moderate** activities refer to activities that take moderate physical effort and make you breathe somewhat harder than normal. Think only about those physical activities that you did for at least 10 minutes at a time.

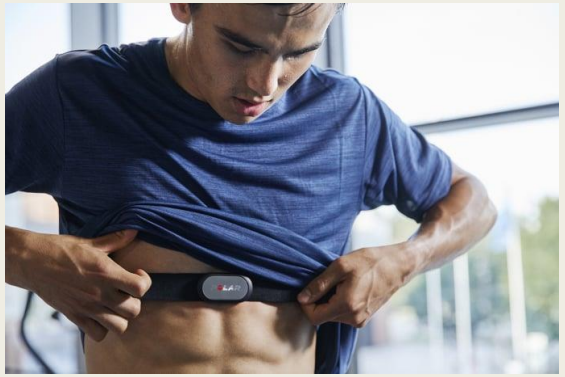
3. During the **last 7 days**, on how many days did you do **moderate** physical activities like carrying light loads, bicycling at a regular pace, or doubles tennis? Do not include walking.

   ____ **days per week**

   ☐ No moderate physical activities → ***Skip to question 5***

SHORT LAST 7 DAYS SELF-ADMINISTERED version of the IPAQ. Revised August 2002.

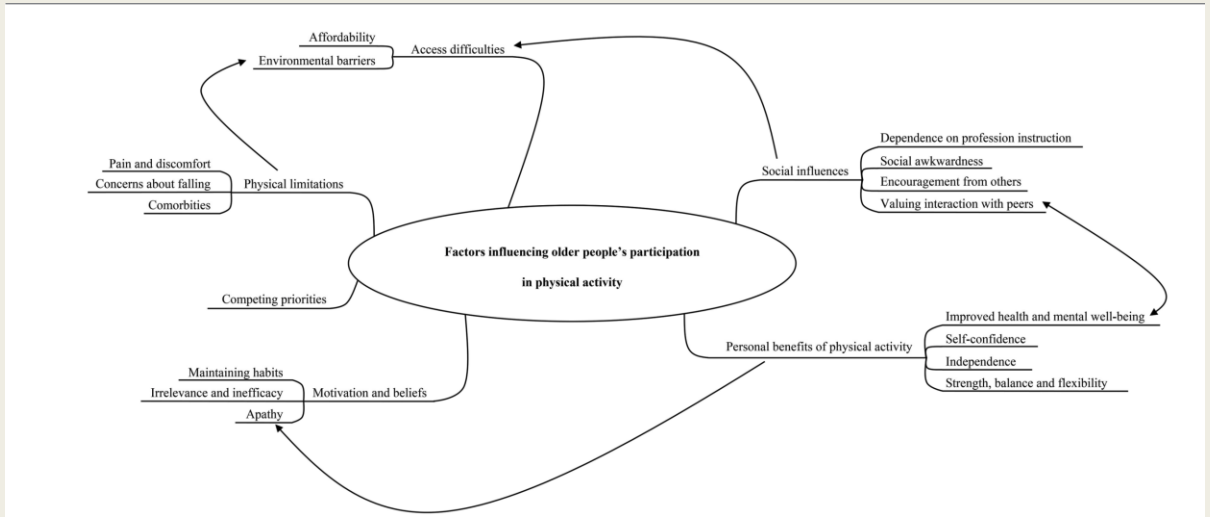






Barriers

- Effectiveness
- Content and delivery
- Capacity and resources
- Reach
- Requirement of multilevel collaboration
- Political support
- Transferability
- Cost
- Sustainability



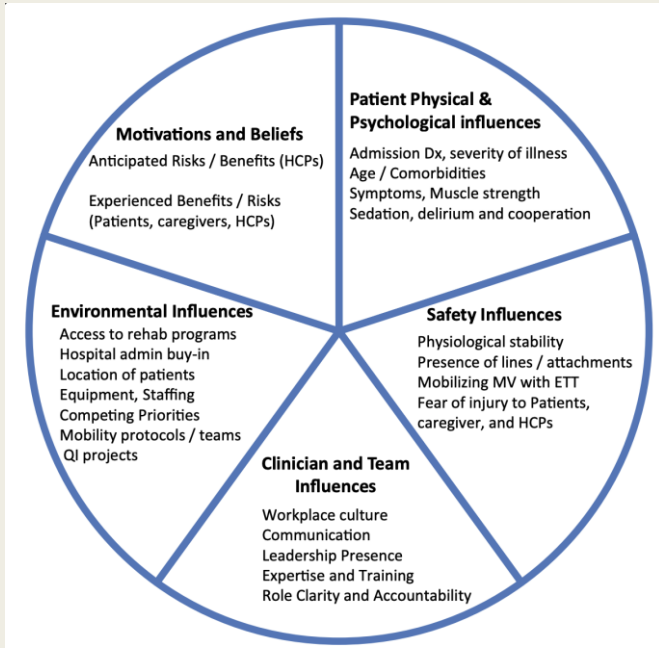


Fig. 2 Summary of findings—themes influencing delivery of physical activity in patients with critical illness. This figure highlights the five themes and 28 sub-themes that were identified in this systematic review. *admin* administrative, *Dx* diagnosis, *ETT* endotracheal tube,





Original Paper

Influence of Pokémon Go on Physical Activity: Study and Implications

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Abstract

Background: Physical activity helps people maintain a healthy weight and reduces the risk for several chronic diseases. Although this knowledge is widely recognized, adults and children in many countries around the world do not get recommended amounts of physical activity. Although many interventions are found to be ineffective at increasing physical activity or reaching inactive populations, there have been anecdotal reports of increased physical activity due to novel mobile games that embed game play in the physical world. The most recent and salient example of such a game is Pokémon Go, which has reportedly reached tens of millions of users in the United States and worldwide.

Objective: The objective of this study was to quantify the impact of Pokémon Go on physical activity.

Methods: We study the effect of Pokémon Go on physical activity through a combination of signals from large-scale corpora of wearable sensor data and search engine logs for 32,000 Microsoft Band users over a period of 3 months. Pokémon Go players are identified through search engine queries and physical activity is measured through accelerometers.

Results: We find that Pokémon Go leads to significant increases in physical activity over a period of 30 days, with particularly engaged users (ie, those making multiple search queries for details about game usage) increasing their activity by 1473 steps a day on average, a more than 25% increase compared with their prior activity level ($P<.001$). In the short time span of the study, we estimate that Pokémon Go has added a total of 144 billion steps to US physical activity. Furthermore, Pokémon Go has been able to increase physical activity across men and women of all ages, weight status, and prior activity levels showing this form of







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